Opinion Pooling V: Geometric and Multiplicative Pooling

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Geometric Pooling

Recap: Linear Pooling

Last time we discussed linear pooling strategies.

- This means you take an average (possibly weighted) of the credences.
- This has a lot of nice features - it preserves unanimity and boundedness, and it only considers individual propositions.
- But fails External Bayesianity; whether you pool first or update first makes a difference.

Today we're looking at a different family of pooling rules based on **geometric means**

Sometimes we express uncertainty as **probabilities**.

But sometimes we express uncertainty as **odds**.

For instance, it might be that A thinks there is a 1 in 10 chance that Michigan will win some competition, while B thinks there is a 1 in 100 chance.

What is half-way between their views?

Linear Pooling

The probabilities are 0.1, and 0.01, so half way between is 0.055, i.e., 5.5%.

Odds Pooling

The odds are 1 in 100, and 1 in 10, so half-way between is 1 in 55, or just under 2%.

Geometric Pooling

It's going to end up at just over 3%, so closer to odds than linear.

Two Answers

What odds pooling and geometric pooling have in common is that they agree on the case where *B* doesn't think Michigan is a 1-in-100 chance, they instead think Michigan is no chance at all.

That is, they have 0 credence, or a 1-in-infinity chance.

Both geometric pooling and odds pooling will say that the combined probability is 0.

We'll come back to this.

The Multi-Stage Process

Geometric pooling works differently from linear pooling:

- **Stage 1**: Choose a partition of logical space. These are distinct ways things could end up that cover all the possibilities.
- **Stage 2**: Pool credences over partition elements using **geometric means**.
- **Stage 3**: Normalize to ensure probabilities sum to 1.
- **Stage 4**: Extend to coarser propositions by summing,

This multi-stage process creates interesting differences from linear pooling.

In general, the geometric mean of n non-negative numbers is given by the following method.

- First **multiply** (not add) the numbers;
- Second, take the n'th root of them. For two numbers that's the square root, for three numbers the cube root, etc.

The geometric mean is always less than or equal to the arithmetic mean (i.e., what you get when you add the numbers and divide by how many numbers there are), with equality only when the numbers are equal.

Example: Anya, Bon, and Carys Return

Consider our epidemiologists again on polio eradication by 2030:

	Eradicated	*Not Eradicated*
Anya	0.2	0.8
Bon	0.6	0.4
Carys	0.7	0.3

Linear pool (equal weights): (0.2 + 0.6 + 0.7)/3 = **0.5**

Geometric Pool: First Attempt

For *Eradicated*, take the geometric mean:

$$0.2^{1/3} \times 0.6^{1/3} \times 0.7^{1/3} \approx 0.438$$

For *Not Eradicated*:

$$0.8^{1/3} \times 0.4^{1/3} \times 0.3^{1/3} \approx 0.458$$

Problem: $0.438 + 0.458 = 0.896 \neq 1$

To get around this, we do something called **normalization**.

Geometric Pool: Normalized

All this means is that we divide each value by the sum so that they'll now sum to 1.[1]

Normalization factor: $\frac{1}{0.438+0.458} \approx 1.118$

Normalized geometric pool:

- *Eradicated*: $0.438 \times 1.118 \approx 0.49$
- *Not Eradicated*: $0.458 \times 1.118 \approx 0.51$

This is close to, but not exactly the same as, the linear pool.

[1] This is something you do in a lot of settings, not always to do with probability, or even to set some designated value to 1, and it's always called normalization.

Given credence functions $Pr_1, \dots, Pr_n$ and partition $\mathcal{S}$:

For each $S \in \mathcal{S}$:

$$\mathsf{GP}_{\mathcal{S}}(Pr_1, \dots, Pr_n)(S) = \frac{(Pr_1(S) \times \cdots \times Pr_n(S))^{1/n}}{\sum_{S' \in \mathcal{S}} (Pr_1(S') \times \cdots \times Pr_n(S'))^{1/n}}$$

For any proposition X that is a disjunction of elements from $\mathcal{S}$:

$$\mathsf{GP}_{\mathcal{S}}(Pr_1, \dots, Pr_n)(X) = \sum_{S \in \mathcal{S}, S \subseteq X} \mathsf{GP}_{\mathcal{S}}(Pr_1, \dots, Pr_n)(S)$$

Partition Sensitivity

A striking feature: **The result depends on which partition you choose**.

Example with Rodrigo and Stefan on rain:

	Heavy Rain	*Light Rain*	*No Rain*
Rodrigo	1/6	1/3	1/2
Stefan	1/3	1/6	1/2

Two Different Partitions

Fine-grained partition: {*Heavy Rain*, *Light Rain*, *No Rain*}

Geometric pool for *No Rain*: **0.52**

Coarse-grained partition: {*Rain*, *No Rain*}

Geometric pool for *No Rain*: **0.5**

Key insight: Same credences, different partitions → different pooled results.

The same example, using the fine-grained partition, shows that geometric pooling violates a bunch of other constraints.

Unanimity Preservation

Both people think rain is 0.5, but the pool does not.

Boundedness

The group credence in rain, 0.48, is below any person's credence.

Eventwise Independence

If Stefan and Rodrigo went to 0 on *Heavy Rain*, and 0.5 on *Light Rain*, the credence in *No Rain* would change, even though no one's credence in *No Rain* would change.

Weighted Geometric Pooling

Just like linear pooling, we can have weighted versions:

Given weights $\Lambda = (\lambda_1, \ldots, \lambda_n)$ that sum to 1:

$$\mathsf{GP}^{\Lambda}_{\mathcal{S}}(P_1, \ldots, P_n)(S) = \frac{\prod_{i=1}^{n} P_i(S)^{\lambda_i}}{\sum_{S' \in \mathcal{S}} \prod_{i=1}^{n} P_i(S')^{\lambda_i}}$$

This allows differential weighting of experts while maintaining the geometric structure. I am **not** going to be discussing this; it's already pushing my mathematical competence.

Why Geometric Pooling?

You might ask: Why use geometric means instead of arithmetic means?

1. It is Externally Bayesian.
2. It maybe gets the right answer in the two detectives case.

But we'll see this comes with costs...

Multiplicative Pooling

Geometric pooling requires weights that sum to 1.

Multiplicative pooling: Allow weights to sum to any positive number.

Most common: weights sum to n (number of individuals).

This creates a whole family of strategies.

Formal Definition: Multiplicative Pooling

Given credences $P_1, \ldots, P_n$, partition $\mathcal{S}$, and weights $\Lambda = (\lambda_1, \ldots, \lambda_n)$:

$$\mathsf{MP}_{\mathcal{S}}^{\Lambda}(P_1, \ldots, P_n)(S) = \frac{\prod_{i=1}^{n} P_i(S)^{\lambda_i}}{\sum_{S' \in \mathcal{S}} \prod_{i=1}^{n} P_i(S')^{\lambda_i}}$$

Note: Same formula as geometric pooling, but weights don't need to sum to 1.

Relationship to Geometric Pooling

Geometric pooling is a special case of multiplicative pooling:

- When $\lambda_1 + \cdots + \lambda_n = 1$: geometric pooling
- When $\lambda_1 + \cdots + \lambda_n \neq 1$: more general multiplicative pooling

Both families share many properties, but differ on some axioms.

Evaluating the Axioms

Let's see how geometric and multiplicative pooling fare on our criteria:

1. Local Unanimity Preservation
2. Global Unanimity Preservation
3. Eventwise Independence
4. Boundedness
5. Non-Dictatorship
6. External Bayesianity
7. Probabilistic Independence Preservation

Do geometric/multiplicative rules satisfy this?

No: Neither satisfies Local Unanimity Preservation.

Recall: Rodrigo and Stefan both assigned credence 0.5 to *No Rain*.

But geometric pool on fine-grained partition gave **0.52**.

What if everyone has exactly the same probability function?

Geometric pooling: Satisfies this

When all $P_i = P$, the geometric pool equals P.

General multiplicative pooling: Violates this

Unless weights sum to 1, even identical credences don't stay put.

Does the pooled credence for X depend only on individual credences for X?

No Multiplicative pooling violates this.

The normalization denominator depends on credences for *all* elements of the partition. So the pooled credence for one proposition is affected by credences in other propositions. We also saw this with the rain example.

(4) Boundedness

Does the pooled credence lie between min and max of individual credences?

No Geometric and multiplicative pooling violate Boundedness.

Example: Rodrigo and Stefan's pooled credence of **0.52** for *No Rain*

Can we avoid giving one person complete control?

Yes Both geometric and multiplicative pooling satisfy this.

As long as multiple individuals get positive weights, no single person dictates.

This parallels linear pooling.

That said, any one person can force the group probability of any number of events to 0 by having credence 0. This is a little dictatorial.

(6) External Bayesianity

If individuals update on evidence, does the group pool commute with updating?

Yes Both geometric and multiplicative pooling are externally Bayesian.

We aren't going to go over the proof of this, maybe we can if there's time and anyone wants, but it's true.

Indeed, *only* multiplicative pooling is externally Bayesian in general.

Why External Bayesianity Matters

This is a major theoretical virtue:

- Group can update on new evidence coherently.
- So there aren't problems of path-dependence.
- The pooling method respects Bayesian reasoning at group level.

This advantage may justify accepting the violations of other axioms.

If individuals judge X and Y independent, does the pool preserve this?

No No non-dictatorial multiplicative rule satisfies this.

The same problem that affects linear pooling affects these rules too.

Most weighted averages (whether arithmetic or geometric) destroy independence.

Comparison and Trade-offs

Axiom	GP
L-Unanimity Preservation	$\boxed{?}$
G-Unanimity Preservation	$\boxed{?}$
Eventwise Independence	$\boxed{?}$
Boundedness	$\boxed{?}$
Non-Dictatorship	$\boxed{?}$
External Bayesianity	$\boxed{?}$
Prob. Independence Pres.	$\boxed{?}$

Score Card: Multiplicative Pooling (General)

Axiom	MP
L-Unanimity Preservation	$\boxed{?}$
G-Unanimity Preservation	$\boxed{?}$
Eventwise Independence	$\boxed{?}$
Boundedness	$\boxed{?}$
Non-Dictatorship	$\boxed{?}$
External Bayesianity	$\boxed{?}$
Prob. Independence Pres.	$\boxed{?}$

General multiplicative pooling is strictly worse than geometric pooling.

Comparing All Classical Methods

Linear pooling: Satisfies Unanimity, Independence, Boundedness

Geometric pooling: Satisfies External Bayesianity, Global Unanimity

Key impossibility: No classical pooling rule satisfies both:

- Boundedness AND External Bayesianity
- Eventwise Independence AND External Bayesianity

When to Use Geometric Pooling?

Geometric pooling might be preferable when:

- It's really important to update frequently, and maintain coherence.
- You want anyone to be able to pull the group credence down by having a low credence.

These considerations depend heavily on the context of use.

When to Use Linear Pooling?

Linear pooling might be preferable when:

- It really matters to keep group credence within the bounds.
- You want to be able to update when you only know credences on this question.
- You won't be updating.
- It's important to interpret the pool as a compromise.

Zeroes

If you don't 100% trust the people who are being pooled, geometric pooling has a risky feature.

Anyone can set the group probability of some event to 0. This seems like a lot of power to give each group member, especially if you aren't sure they are right.

In practice, I'd be very hesitant to use geometric pooling without first eliminating outliers, just for this reason.

This fact about zeroes relates to the issue about partition sensitivity.

The more fine-grained the partition you use, the more chance someone will have a credence 0 in one or other of the cells.

So you can very easily end up with some realistic scenario getting group credence 0, because every particular way that scenario might come about gets credence 0 according to some group member or other.

Key theoretical results:

- **No classical pooling rule satisfies all the intuitive axioms**.
- So on any occasion we have to make choices about which axioms are most important.
- You have to be sensitive to why you are pooling to make this choice.

Philosophical Implications

Partition Sensitivity Revisited

Geometric/multiplicative pooling's partition-sensitivity raises some questions:

1. What procedure should we use for working out the relevant partition?
2. Is there a correct partition for a particular use? If so, what features (pragmatic or theoretical) make it correct?

Like with the pool-then-update or update-then-pool choice, this adds another choice that's prior to being able to pool.

External Bayesianity and Justification

Recall that Lackey's GEAA account requires:

- A group belief is justified only if it based on justified individual beliefs, and
- To retain rationality, a group must update appropriately on new evidence.

It feels very hard to do this without external Bayesianity.

- Ensures group can update coherently
- Maintains link between individual and group justification
- May be essential for justified group belief over time

The Expertise Problem

We'll mostly use the unweighted (or equal weight) versions, but obviously if you use weighted versions, these also raise pre-pooling questions about how to assign weights.

And in geometric pooling it's even harder because the holism of the process means it is harder to say "I'm an expert on this and you're an expert on that."

A view one could have, although I don't know if anyone does have it, is that in cases of peer disagreement, the peers should:

1. All converge on a pooled credence, just like the standard Equal Weight View; but
2. This pooled credence should be given by geometric pooling.

This will imply that people with low credence get to really influence the overall result.

Looking Ahead

Next Steps in the Course

Moving forward, we'll explore how we might go about choosing between these in particular contexts, how they relate to the motivations for pooling, and (perhaps) whether there are any further options we left out.

For now, here's a summary

1. Pooling involves trade-offs.
2. No single pooling method is best for all purposes.
3. External Bayesianity is geometric/multiplicative pooling's key virtue.
4. But it comes at the cost of other desirable properties, such as unanimity.
5. So we need to think about which of these we care most about on an occasion.